DSE Bullet Series – Network Fundamentals (Solutions 05)

Name: _____ ( ) Class:____ Date: _____

1. Past Questions (2018–2022) #LAN/WAN first, then the devices

Year	Past Question	Focus / Marks
2018	Functions of a NIC, network cable and switch	Connect to a network; transmit signals; forward data in a LAN
2019	Wired vs wireless; packets and IP delivery	Stability/interference; segmentation, routing, reassembly
2021	Home networks X/Z; uses of a network	Router/access point; resource sharing and communication
2022	Network devices P/Q; router and DNS	Switch/access point; connect networks; domain name to IP

2. Key Notes #A network is more than one cable

LAN and WAN: a LAN covers a school, office or home and usually has higher speed and lower cost; a WAN covers a city or country, often uses ISP lines, and normally has higher cost and lower speed. A network shares hardware, software and data, and supports communication.

Network hardware: a network interface card (NIC) connects a device to a network; UTP cable and optical fibre carry signals; a switch forwards frames within the same LAN; a router connects different networks and chooses a path using IP addresses; an access point (AP) provides wireless access; a modem converts signals for connection to an ISP.

Wired and wireless: wired connections are usually more stable, faster and less affected by interference, but are restricted by cables; wireless connections allow mobility but are affected by distance, walls, interference and shared bandwidth. Roaming lets a device stay connected while moving between APs.

IPv4 and IPv6 are versions of the Internet Protocol. An IPv4 address has four groups of numbers; IPv6 addresses are longer and provide a much larger address space. TCP/IP segments data into packets, adds source and destination IP addresses, sends them through routers, and uses TCP to check and reassemble them.

URL, DNS, HTTP and HTTPS: a URL is the address of a resource; DNS translates a domain name into an IP address; HTTP transfers web pages; HTTPS is HTTP protected by encryption.
*Common mistakes: confusing a switch with a router; treating an access point as a router; writing only ‘fast’ or ‘convenient’; saying DNS is a web server; describing packet transfer without segmentation, routing and reassembly.

3. Similar Questions #Same idea, different scenario

1. A school library adds a wireless access point to its existing wired network.

(a) State **two** differences between wired and wireless connections.

【Answer】 Wired is usually more stable, faster and less affected by interference; wireless allows mobility but is affected by distance, walls and interference. Wired is restricted by cables, while wireless can support multiple mobile devices.

(b) Explain the purpose of the access point in this network.

【Answer】 The AP connects wireless devices to the wired LAN; users can access network resources and the Internet within signal range; multiple devices can share the network connection.

2. Ka-fai sets up a home network with a broadband modem, a router and an access point.

(a) State one function of the router and one function of the access point.

【Answer】 The router connects the home LAN to the Internet and forwards packets using IP addresses; the AP provides wireless access so phones and tablets can connect to the LAN.

(b) Explain why typing a domain name in a browser can reach a website.

【Answer】 DNS translates the domain name into the web server's IP address; the browser then sends an HTTP/HTTPS request to that IP address and receives the web page.

3. Mei sends a large file from computer A to computer B on the Internet.

(a) Describe how the file is transferred from A to B.

【Answer】 The file is divided into packets; each packet has source and destination IP addresses; routers forward the packets according to the destination address; B reassembles the packets in the correct order.

(b) How does TCP help ensure that the file is complete?

【Answer】 TCP uses acknowledgements; if a packet is lost or damaged, the sender retransmits it; the receiver reassembles the packets in sequence to reconstruct the file.

DSE Bullet Series – Internet Services and Applications (Solutions 06)

1. Past Questions (2020–2025) #Search, formats and streaming

Year	Past Question	Focus / Marks
2020	MP4 vs AVI; technical problems in streaming	Compression and compatibility; bandwidth, delay, buffering
2023	Verifying online information; poor streaming	Author/date/cross-check; congestion and packet loss
2024	Sending email in HTML rather than plain text	Layout, images, hyperlinks
2025	Email attachment vs hyperlink; HTML vs PDF	File size and broken links; cross-platform layout

2. Key Notes #Search, formats and streaming

Search strategies: clarify the information needed, then choose keywords; use quotation marks for an exact phrase, OR to broaden a search, a minus sign to exclude a word, and site: to limit the search to a website. You can also filter results by image, video or news source.

Verifying online information: reliability means consistency with other sources and clear citation; credibility means the author or organisation has relevant qualifications; currency means the information is up to date. Cross-check several sources, trace the original source, and check the domain name and date.

Web multimedia formats: JPEG suits photographs and gives smaller files; PNG is lossless and supports transparency; GIF supports animation but is limited to 256 colours; SVG is a vector format that remains sharp when enlarged. MP3 audio is small and widely supported; AAC usually gives better quality; FLAC is lossless but large. MP4 video is relatively small, compatible and suitable for streaming.

Online services: email uses To, Cc and Bcc to control recipients; large files can be sent as attachments, embedded items or hyperlinks. Instant messaging supports text, voice and images; forums organise discussions by topic; social networks share updates; files can be transferred through cloud storage or P2P; remote login controls another device from a distance.

Streaming: data is played while it is being downloaded, so the user does not wait for the whole file. This shortens waiting time and saves local storage, but it needs sufficient and stable bandwidth; otherwise buffering, pauses and lower quality occur. Quality depends on bandwidth, compression/codec, bit rate, resolution, frame rate and network stability.

IoT and cloud: IoT connects sensors and devices to collect data; cloud services provide storage and computing. Together they support smart-city applications such as transport, environmental monitoring and energy management.

*Common mistakes: writing only ‘search the Internet’ without a strategy; treating popularity as reliability; confusing JPEG and PNG; describing streaming only as ‘fast’; reversing the interaction direction of a video conference and a webcast.

3. Similar Questions #Same idea, different scenario

1. Mei finds a blog post claiming that a health product improves eyesight within one week.

(a) Suggest **two** ways to verify whether the article is reliable.

【Answer】Check the author's or organisation's qualifications; check the publication date and whether it is current; compare it with reliable sources such as government, university or professional bodies; trace the original research or data cited.

(b) Explain why drawing a conclusion from one website is risky.

【Answer】Anyone can publish content online and it may not be reviewed; a single source may be outdated, incomplete, wrong or commercially biased; without corroboration, users can be misled.

2. Ka-fai uploads a video of a school event for students to watch online, but the video frequently pauses for some students.

(a) Suggest a suitable video format for streaming and give one reason.

【Answer】MP4: it usually gives smaller files with efficient compression and is widely supported by browsers and devices, making it suitable for streaming; AVI files are often larger and less compatible.

(b) Apart from the file format, state **two** technical factors that may cause poor streaming.

【Answer】Insufficient or unstable bandwidth; network congestion or an overloaded server; packet loss or delay; a bit rate or resolution that is too high; Wi-Fi interference.

3. About 2,000 parents watch a school anniversary ceremony online, but they do not need to speak.

(a) Should the school use a video conference or a webcast? Explain.

【Answer】A webcast: it is mainly one-way and can reach a large audience; a video conference is usually designed for two-way interaction among a smaller number of participants.

(b) Some parents want to watch the ceremony later. Suggest one technology and one benefit.

【Answer】Record the ceremony and provide it as a streamed video: parents can replay, pause and choose segments at any time; streaming lets playback begin without downloading the whole file, saving time and local storage.

DSE Bullet Series – Online Threats and Security (Solutions 07)

1. Past Questions (2012–2022) #Identify the threat, then the measure

Year	Past Question	Focus / Marks
2012	Unsecured Wi-Fi; Bluetooth virus; public/private keys	Wi-Fi encryption; antivirus; private-key decryption
2014	Sending a file to a branch using PKI	Public-key encryption; private-key decryption; prevent interception
2016	GovWiFi unencrypted connection; fingerprint access control	Interception risk; fingerprints harder to copy
2017	Website verifies a customer with a security token	One-time password; time limit; reduce password theft
2018	Can PKI prevent eavesdropping and interception?	Ciphertext is unreadable; verify sender and integrity
2021	Public Wi-Fi, phone auto-lock and antivirus	Avoid sensitive transactions; reduce theft and infection
2022	Mobile app permissions and personal data	Grant only necessary permissions; privacy and leakage

2. Key Notes #Threats, authentication, encryption

Threats: a virus attaches to files; a worm copies itself and spreads through a network; a Trojan disguises itself as normal software; spyware steals information; ransomware encrypts files. Other threats include unauthorised access, interception, phishing, spam and DoS attacks.

Defences: antivirus software scans for malware; a firewall filters network traffic by rules; system updates patch vulnerabilities; browsers block dangerous sites; WPA encrypts wireless traffic; a VPN creates an encrypted tunnel and reduces interception risk on public Wi-Fi. A firewall is not antivirus software.

Authentication and authorisation: authentication proves who you are, using a password, biometrics, smart card, security token or SMS code; authorisation decides what you may do, following the least-privilege principle. Two-factor authentication combines two methods.

Encryption: plaintext is converted into ciphertext and can be restored only with the correct key. Public-key encryption and private-key decryption protect confidentiality; a digital signature is created with the sender's private key and verified with the sender's public key to prove origin and integrity. PKI and digital certificates bind an identity to a public key; longer keys are harder to break.

E-commerce and privacy: HTTPS/SSL encrypts and checks a digital certificate; use an SMS code or security token for authentication; avoid transactions on public computers or unencrypted Wi-Fi; log out afterwards.

Apps should receive only necessary permissions; personal data should be collected for a stated purpose and protected by encryption.

Use C-F-B: Context (C) + security feature (F) + benefit (B). Example: online banking uses HTTPS (C), SSL encrypts transaction data (F), so intercepted data cannot be read (B).
*Common mistakes: writing ‘encrypt’ without naming the key; confusing authentication with authorisation; assuming HTTPS means a website is trustworthy; treating a firewall as antivirus software; writing only ‘be careful’.

3. Similar Questions #Same idea, different scenario

1. Mei connects to an open public Wi-Fi network in a coffee shop and is about to log in to online banking.

(a) State **two** risks of making an online transaction on that Wi-Fi network.

【Answer】 The public Wi-Fi may be unencrypted, so transaction data can be intercepted; an attacker may set up a fake hotspot to steal the bank password or SMS code; other devices on the same network may spread malware.

(b) Suggest **two** ways to reduce the risks.

【Answer】 Use mobile data or a VPN; check that the website uses HTTPS and verify its digital certificate; avoid sensitive transactions on public Wi-Fi; enable two-factor authentication and log out afterwards.

2. Ka-fai receives a bank email with 'account.pdf.exe' asking him to log in at once.

(a) Identify the type of threat and explain **two** suspicious features.

【Answer】 Phishing. The sender pretends to be a bank and creates urgency; the double extension '.pdf.exe' disguises an executable file that may be a Trojan or ransomware; a link may lead to a fake website that steals account details.

(b) Suggest **three** security measures.

【Answer】 Do not open the attachment or click the link; verify through the bank's official website or hotline; scan with antivirus software; update the system and software; back up important files; mark the message as spam.

3. A school sends student data from the main campus to a branch campus using PKI.

(a) Explain the roles of the public key and private key in this transfer.

【Answer】 The sender encrypts the file with the branch's public key; only the branch's private key can decrypt it, protecting confidentiality. The sender also creates a digital signature with their own private key; the branch verifies it with the sender's public key to check origin and integrity.

(b) Explain how a digital certificate helps the branch confirm the sender's identity.

【Answer】 The certificate is issued by a trusted certificate authority and binds the sender's identity to their public key; if the certificate and digital signature are valid, the branch can confirm the sender's identity and that the file has not been altered.

DSE Bullet Series – Web Authoring and HTML (Solutions 15)

1. Past Questions (2012–2025) #HTML is markup, not programming

Year	Past Question	Focus / Marks
2012	Hyperlinks to different files; browser behaviour	href, file format, opening method
2016	Settings for inserting a table into a web page	Rows, columns, table headings, cells
Practice Paper	Frame, background colour, font colour	frame, bgcolor, colour code
Sample Paper	Traditional/simplified Chinese pages; steps in web authoring	Unicode/character set; HTML does not need compilation

2. Key Notes #Know the tag, attribute and file

HTML is a markup language, not a programming language; a browser reads the markup and displays the page, so no compilation is needed.

Basic structure: <html>, <head> (contains <title>) and <body> (contains visible content).

Common tags: <h1>–<h6> headings, <p> paragraphs,
 line break, <hr> horizontal rule, and /<ol>/ lists.

A hyperlink uses text; an image uses <img src="file" alt="alternative text">. Each path must point to the correct HTML or image file.

A table uses <table>, <tr> for a row, <th> for a heading and <td> for a cell; do not use spaces to lay out a table.

Multimedia: JPG/PNG for images, MP3 for audio and MP4 for video. Consider file size and loading time before adding a file.

Site organisation: a home page, navigation, internal links and consistent headings. A Chinese page can use Unicode (UTF-8) to display traditional and simplified Chinese together.

A frame divides the browser window into several areas, each loading a different HTML file; a framed site therefore needs more than one HTML file.

*Common mistakes: treating HTML as a compiled program; using
 as a paragraph; omitting alt; using a wrong image path; designing only the appearance without navigation.

3. Tag Example #Read from the outside in

```
<html>
<head><title>School Open Day</title></head>
<body>
<h1>School Open Day</h1>
<a href="map.html">Campus map</a>
<img src="campus.jpg" alt="Campus photo">
</body>
</html>
```

The browser reads page information from <head>, then displays the heading, text, hyperlink and image inside <body>.

4. Similar Questions #A different page, the same pattern

1. Hiu-yan designs a School Open Day web page. Part of the code is shown below:

```
<h1>School Open Day</h1><p>Parents and students are welcome.</p>
<a href="map.html">Campus map</a><img src="campus.jpg" alt="Campus">
```

(a) Write the complete tag that takes the user to map.html.

【Answer】Campus map

(b) Explain the purpose of the alt attribute.

【Answer】It displays alternative text when the image cannot be shown; it also lets a screen reader describe the image to a visually impaired user.

(c) Apart from an image, name one multimedia element that can be added to a web page and give a suitable file format.

【Answer】Video: MP4; or audio: MP3. (Other reasonable answers are acceptable.)

2. Ka-fai uses a table to present extracurricular activities on a web page:

```
<table><tr><th>Activity</th><th>Date</th></tr>
<tr><td>Singing Contest</td><td>1 May</td></tr></table>
```

(a) State the tags used for a table heading and a table row in the code above.

【Answer】Table heading: <th>; table row: <tr>.

(b) Explain why a web page should not turn all its text into images.

【Answer】Text in an image cannot be selected, searched or read by a screen reader; image files are usually larger and load more slowly; the text is also harder to edit and adapt to different screen sizes.

3. The school is making a web page for mobile phones that contains both traditional and simplified Chinese. Suggest a character encoding and explain why, and state one design consideration for this web page.

【Answer】Unicode (UTF-8): the character set can represent both traditional and simplified Chinese. Also provide clear navigation, add alt text to images, compress images, and use fonts and button sizes suitable for mobile phones. (Other reasonable answers are acceptable.)